Marc Benedí

marc.benedi@tum.de | https://marcb.pro | marcbenedi | marcbenedi

Education

Visual Computing Group - Technische Universität München
Munich, DE

PH.D., COMPUTER VISION & DEEP LEARNING — *January 2023 - Present*

- Researching the generation of human avatars from monocular depth observations, aiming to advance the capabilities of real-time interactive systems and enhance user immersion in digital environments.
- Teach graduate-level lectures and seminars on topics related to computer vision and machine learning.
- Supervise undergraduate and graduate student projects, providing guidance and expertise to support innovative research and practical applications in computer vision.

Technische Universität München
Munich, DE

M.SC., COMPUTER SCIENCE — *October 2019 - September 2022*

- Specialised in Computer Vision, Machine Learning and Deep Learning.
- Experience in CUDA/GPU programming with focus on high performance computing.
- Thesis: ”Learning Robust Correspondences for Relative Camera Pose Estimation”.

University of Edinburgh
Edinburgh, UK

ERASMUS+ (EXCHANGE) — *January 2018 - May 2018*

- Focus on Artificial Intelligence and Advanced Software Engineering lectures.

Universitat Politècnica de Catalunya, Barcelona Tech
Barcelona, ES

B.SC., COMPUTER ENGINEERING — *September 2014 - July 2018*

- Specialised in Computer Science.
- Thesis: ”Design of an environment for solving pseudo-Boolean optimization problems”.

Experience

Visual Computing Group - Technische Universität München
Munich, DE

RESEARCH FELLOWSHIP — *May 2021 - April 2022*

- Fellowship awarded to outstanding Master students to allow them developing state-of-the-art models.
- Developed a Convolutional Neural Network (CNN) for estimating correspondences and perform robust optimization to reconstruct 3D rigid-motions.

Visual Computing Group - Technische Universität München
Munich, DE

DEVOPS/SYSTEMS ENGINEER — *April 2020 - September 2022*

- Re-designed the laboratory’s infrastructure into a high-performance SLURM cluster, specifically tailored for advanced visual computing and AI research, significantly optimizing task scheduling and resource allocation.
- Designed and implemented a full-stack platform for managing the cluster’s scheduling, storage, and node resources, using Vue.js, Python and Docker for containerization.
- Deployed and sustained key infrastructure technologies such as LDAP, NFS, and PXE, improving system integrity and performance, and ensuring high availability and scalability of computing resources critical to research continuity.

WARP - Autonomus Driving
Munich, DE

COMPUTER VISION INTERN — *April 2020 - September 2020*

- Implemented Simultaneous Localization and Mapping (SLAM) for an autonomous vehicle, integrating sensor odometry and LIDAR-generated point clouds to enhance navigation precision. Utilized C++, Point Cloud Library (PCL), Robot Operating System (ROS), Ceres Solver, and Google Cartographer for robust and efficient mapping and localization.

Helm Mobile
Barcelona, ES

SOFTWARE ENGINEER — *September 2018 - September 2019*

- Developed Android applications leveraging Kotlin, Dagger for dependency injection, and Retrofit for API integration; engineered backend systems using Kotlin and SQL; and designed web interfaces with Vue.js and JavaScript—all with a strong focus on coding standards and Clean Architecture to ensure high-quality, maintainable, and scalable code.

CERN - European Organization for Nuclear Research
Geneva, CH

INTERNSHIP - ”SUMMER STUDENT” — *June 2018 - September 2018*

- Engineered and deployed a Kubernetes cluster for the BE-OP-LHC team at CERN, enhancing system scalability, availability, and response time by 118%.
- Developed Ansible scripts to automate cluster deployment across various machine groups, ensuring consistency and efficiency in operations.

Universitat Politècnica de Catalunya, Barcelona Tech
Barcelona, ES

RESEARCH FELLOWSHIP *September 2017 - July 2018*

- Derived a novel conversion for transforming Binary Decision Trees into Conjunctive Normal Forms, enhancing the efficiency of Boolean Satisfiability Problem (SAT) representations and solutions.

Ernst & Young
Barcelona, ES

SOFTWARE ENGINEER INTERNSHIP *June 2017 - September 2017*

- Developed responsive and dynamic web applications by creating the front-end using HTML, CSS, JavaScript, and jQuery, and building the back-end system in Java.

Projects

Learning Robust Correspondences Estimation

TECHNICAL UNIVERSITY OF MUNICH *2022*

- An end-to-end model for pose estimation between pairs of RGB-D frames. The key idea is learning confidence scores for correspondences in a self-supervised manner, helping to filter outliers and improve pose accuracy.

End-to-end Sparse Correspondence Prediction for Relative Camera Pose Estimation

TECHNICAL UNIVERSITY OF MUNICH *2021*

- A method for end-to-end learnable, differentiable relative pose registration of RGB-D frames. The method predicts sparse correspondences using probability heatmaps, which then are fed into differentiable weighted Procrustes to estimate the final transformation between the frames.

KinectFusion

TECHNICAL UNIVERSITY OF MUNICH *2021*

- Implemented "KinectFusion: Real-Time Dense Surface Mapping and Tracking" for real-time reconstruction of large-scale indoor scenes from RGB-D frame using C++ and CUDA for high-performance.

Divergence-Free Shape Correspondence with Time Dependent Vector Fields

TECHNICAL UNIVERSITY OF MUNICH *2020*

- Extended the paper "Divergence-Free Shape Correspondence by Deformation" to represent sequences of non-rigid deformations as a time dependent vector field.

Awards

Distinctions

- **Technical University of Munich** - 3D Scanning & Motion Capture (2020)
- **Universitat Politècnica de Catalunya, Barcelona Tech** - Computer Security (2017), Software Architecture (2017), Operative Systems (2015)
- **University of Edinburgh** - Reasoning Agents (2018), Software Testing (2018)

Scholarships

- **Research Scholarship**, Munich 2021-2022 - Fellowship awarded to outstanding Master students by the Visual Computing Group @ TUM
- **Research Scholarship**, Barcelona 2017-2018 - Research fellowship awarded by the Spanish Government
- **Academic Scholarship**, Barcelona 2014-2018 - Scholarship awarded by the Spanish Government for the whole duration of the bachelors

Hackathons

- **HackUPC**, Barcelona 2017 - GFI Sponsor prize
- **FoodHacks**, Berlin 2016 - First prize, Lidl Sponsor prize
- **Stirhack**, Stirling 2016 - First prize

Others

For more information, please visit my website at: https://marcb.pro

Languages

- **English**, C1 (Fluent)
- **Spanish**, Native
- **German**, A1 (Basic)
- **Catalan**, Native